№3

$$\xi \sim p(x) = \begin{cases} e^{-x/\theta}/\theta, & x \geq 0 \\ 0, & x < 0 \end{cases} \quad \theta > 0 \quad \overset{\shortparallel}{\Theta} \qquad n = 3 \quad \vec{x}_n$$

$$\tilde{\Theta}_1 = \bar{x} \qquad \tilde{\Theta}_2 = x_{(2)}$$

а) Проверить оценки на несмещ.

$$M\xi = \int_{-\infty}^{\infty} x\, dF(x) = \int_0^{\infty} \frac{x}{\theta} e^{-\frac{x}{\theta}} dx = \frac{-\theta}{\theta} \int_0^{\infty} x\, de^{-\frac{x}{\theta}} = \cancel{-xe^{-\frac{x}{\theta}}\Big|_0^{\infty}} +$$

$$+ \int_0^{\infty} e^{-\frac{x}{\theta}} dx = -\theta e^{-\frac{x}{\theta}}\Big|_0^{\infty} = \theta$$

$$M\xi^2 = \int_{-\infty}^{\infty} x^2 dF(x) = \frac{1}{\theta}\int_0^{\infty} x^2 e^{-\frac{x}{\theta}} dx = -\int_0^{\infty} x^2 de^{-\frac{x}{\theta}} = \cancel{-x^2 e^{-\frac{x}{\theta}}\Big|_0^{\infty}} +$$

$$+ 2\underbrace{\int_0^{\infty} x e^{-\frac{x}{\theta}} dx}_{M\xi \cdot \theta} = 2M\xi \cdot \theta = 2\theta^2 \Rightarrow D\xi = M\xi^2 - (M\xi)^2 = 2\theta^2 - \theta^2 = \theta^2$$

$$\Rightarrow M\xi = \theta \quad D\xi = \theta^2$$

$\tilde{\Theta}_1 = \bar{x}$

несм.: $M[\tilde{\Theta}_1] \underset{\text{надо}}{=} \theta$: $M\bar{x} = M\left[\frac{1}{n}\sum_{i=1}^{n} x_i\right] = \frac{1}{n} nM\xi = \theta \Rightarrow \tilde{\Theta}_1$ - несм.

$\tilde{\Theta}_2 = x_{(2)}$

$$\xi_{(k)} \sim \sum_{i=k}^{n} C_n^i F^i(t)(1-F(t))^{n-i} \qquad k = 2, \; n = 3$$

$$\xi_{(2)} \sim C_3^2 F^2(t)(1-F(t)) + C_3^3 F^3(t) = 3F^2(t)(1-F(t)) + F^3(t) =$$

$$= 3F^2(t) - 2F^3(t) = \Psi(t) \qquad F(t) = (1 - e^{-\frac{t}{\theta}})$$

$$p(t) = e^{-t/\theta}/\theta$$

$$\psi(t) = (\Psi(t))' = 6F(t)p(t) - 6F^2(t)p(t)$$

$$\psi(x) = 6\left((1-e^{-\frac{x}{\theta}})\frac{e^{-\frac{x}{\theta}}}{\theta}\right) - 6\left((1-e^{-\frac{x}{\theta}})^2 \frac{e^{-\frac{x}{\theta}}}{\theta}\right) = 6\left(\cancel{\frac{e^{-\frac{x}{\theta}}}{\theta}} - \frac{e^{-\frac{2x}{\theta}}}{\theta} -\right.$$

$$\left. - \cancel{\frac{e^{-\frac{x}{\theta}}}{\theta}} + \frac{2e^{-\frac{2x}{\theta}}}{\theta} - \frac{e^{-\frac{3x}{\theta}}}{\theta}\right) = 6\left(\frac{e^{-\frac{2x}{\theta}}}{\theta} - \frac{e^{-\frac{3x}{\theta}}}{\theta}\right)$$

несм.: $M[\tilde{\Theta}_2] \underset{\text{надо}}{=} \theta$ $\quad M x_{(2)} = \underbrace{6\int_0^{\infty} x \frac{e^{-\frac{2x}{\theta}}}{\theta} dx}_{I_1} - \underbrace{6\int_0^{\infty} x \frac{e^{-\frac{3x}{\theta}}}{\theta} dx}_{I_2}$

$$I_1 = \cancel{-3xe^{-\frac{2x}{\theta}}\Big|_0^\infty} + 3\int_0^\infty e^{-\frac{2x}{\theta}}dx = -\frac{3\theta}{2}e^{-\frac{2x}{\theta}}\Big|_0^\infty = \frac{3}{2}\theta$$

$$I_2 = -2xe^{-\frac{3x}{\theta}}\Big|_0^\infty + 2\int_0^\infty e^{-\frac{3x}{\theta}}dx = -\frac{2\theta}{3}e^{-\frac{3x}{\theta}}\Big|_0^\infty = \frac{2}{3}\theta$$

$\Rightarrow M\tilde{\theta}_2 = I_1 - I_2 = \frac{3}{2}\theta - \frac{2}{3}\theta = \frac{5}{6}\theta \neq \theta \Rightarrow \tilde{\theta}_2$ – <u>смещ.</u>

<u>исправим</u>: $\tilde{\theta}_2' = \frac{6}{5}x_{(2)}$ $\quad M\tilde{\theta}_2' = \theta \Rightarrow \tilde{\theta}_2'$ – <u>несм.</u>

в) $\tilde{\theta}_1 = \bar{x}$ $\quad D\tilde{\theta}_1 = D\bar{x} = D\left(\frac{1}{n}\sum_{i=1}^n x_i\right) = \frac{D\xi}{n} = \frac{\theta^2}{n} = \frac{\theta^2}{3}$

$\tilde{\theta}_2' = \frac{6}{5}x_{(2)}$ $\quad D\tilde{\theta}_2' = D\left(\frac{6}{5}x_{(2)}\right) = \frac{36}{25}Dx_{(2)}$

$Dx_{(2)} = M[\tilde{\theta}_2^{\,2}] - (M[\tilde{\theta}_2])^2$

$$M\tilde{\theta}_2^{\,2} = 6\int_0^\infty x^2\left(\frac{e^{-\frac{2x}{\theta}}}{\theta} - \frac{e^{-\frac{3x}{\theta}}}{\theta}\right)dx = \underbrace{6\int_0^\infty x^2\frac{e^{-\frac{2x}{\theta}}}{\theta}dx}_{I_1'} - \underbrace{6\int_0^\infty x^2\frac{e^{-\frac{3x}{\theta}}}{\theta}dx}_{I_2'}$$

$$I_1' = -3\int_0^\infty x^2 de^{-\frac{2x}{\theta}} = \cancel{-3x^2e^{-\frac{2x}{\theta}}\Big|_0^\infty} + \underbrace{6\int_0^\infty xe^{-\frac{2x}{\theta}}dx}_{I_1\cdot\theta} = \frac{3}{2}\theta^2$$

$$I_2' = -2\int_0^\infty x^2 de^{-\frac{3x}{\theta}} = -2x^2e^{-\frac{3x}{\theta}}\Big|_0^\infty + \underbrace{4\int_0^\infty xe^{-\frac{3x}{\theta}}dx}_{I_1\frac{2\theta}{3}} = \frac{4}{9}\theta^2$$

$\Rightarrow M\tilde{\theta}_2^{\,2} = \frac{3}{2}\theta^2 - \frac{4}{9}\theta^2 = \frac{19\theta^2}{18}$

$\Rightarrow Dx_{(2)} = \frac{19}{18}\theta^2 - \frac{25}{36}\theta^2 = \frac{13}{36}\theta^2$

$\Rightarrow D\tilde{\theta}_2' = \frac{36}{25}\cdot Dx_{(2)} = \frac{36}{25}\cdot\frac{13}{36}\theta^2 = \frac{13}{25}\theta^2$

$D\tilde{\theta}_1 = \frac{\theta^2}{3}$ $\quad D\tilde{\theta}_1 < D\tilde{\theta}_2' \Rightarrow \tilde{\theta}_1$ – <u>эффективнее</u> $\tilde{\theta}_2'$

$\forall\theta > 0$

$D\tilde{\theta}_2' = \frac{13}{25}\theta^2$

с) Исследовать оц. на эффективность с помощью нер-ва Крамера-Рао

для того, чтобы восп. нер-вом К.-Р., нужны рег. модель и оценка

1) модель — рег.

а) $p(x,\theta)=\frac{e^{-\frac{x}{\theta}}}{\theta}$ $\{(0;+\infty)\}$ — непр. дифф. на $\theta>0$

б) $\frac{\partial}{\partial\theta}\int dF=\int d\frac{\partial F}{\partial\theta}\to\int_0^\infty\frac{\partial p(x,\theta)}{\partial\theta}dx=0$

$$\int_0^\infty\frac{\partial}{\partial\theta}\left(\frac{e^{-\frac{x}{\theta}}}{\theta}\right)dx=\int_0^\infty\frac{xe^{-\frac{x}{\theta}}}{\theta^3}dx-\int_0^\infty\frac{e^{-\frac{x}{\theta}}}{\theta^2}dx=\frac{1}{\theta}-\frac{1}{\theta}=0$$

в) $I(\theta)=M\left[\left(\frac{\partial\ln p(x,\theta)}{\partial\theta}\right)^2\right]=\int_0^\infty\left(\frac{x}{\theta^2}-\frac{1}{\theta}\right)^2\frac{e^{-\frac{x}{\theta}}}{\theta}dx=\int_0^\infty\frac{x^2}{\theta^4}\frac{e^{-\frac{x}{\theta}}}{\theta}dx-$

$$-2\int_0^\infty\frac{x}{\theta^3}\frac{e^{-\frac{x}{\theta}}}{\theta}dx+\int_0^\infty\frac{1}{\theta^2}\frac{e^{-\frac{x}{\theta}}}{\theta}dx=\frac{2}{\theta^2}-\frac{2}{\theta^2}+\frac{1}{\theta^2}=\frac{1}{\theta^2}$$

$\Rightarrow I(\theta)$ — непр., >0 на $\theta>0$

$\Rightarrow$ модель — регулярна (1)

2) оц. — регулярна

Найдём по дост. усл. Рассмотрим $\tilde\theta_1$

1) оц. несм. —

2) модель рег. —

3) $D\tilde\theta_1=\frac{\theta^2}{3}$ — огр. на $\forall$ конт. из $\theta>0$

дост. усл.

$\Rightarrow\tilde\theta_1$ — регулярная оц. (2)

из (1) и (2) $\Rightarrow$ Нер-во Крамера-Рао работает

$\Rightarrow D\tilde\theta_1=\frac{\theta^2}{3}\ ?\ \frac{1}{\underset{\underset{3}{\parallel}}{n}\frac{1}{\theta^2}}\Rightarrow\frac{\theta^2}{3}=\frac{\theta^2}{3}$ дост. усл. $\Rightarrow$

оц. $\tilde\theta_1$ — эффективна

(см. след. стр.)

Рассмотрим $\tilde{\theta}_2'$

а) оц. $\tilde{\theta}_2'$ – несм. ✓

б) модель рег. – ✓

в) $D\tilde{\theta}_2' = \frac{13}{25}\theta_2'$ – огр. на $\forall$ комп. из $\theta>0$ ✓

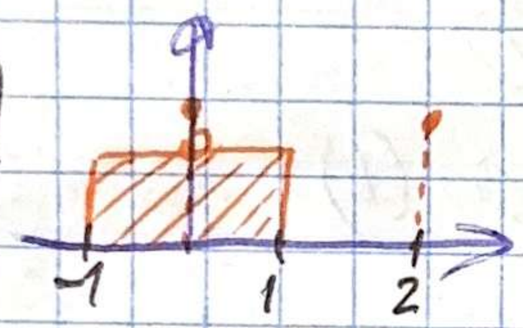
дост. усл. $\Rightarrow \tilde{\theta}_2'$ – рег. оц. ✓ (2')

из (1) и (2') $\Longrightarrow$ работает нер-во Кр.-Р. $\Rightarrow$

$D\tilde{\theta}_2' = \frac{13}{25}\theta^2 > \frac{\theta^2}{3} \Rightarrow$ ничего сказать об эфф. нельзя

но по (T) ! эфф. оц. $\Rightarrow \tilde{\theta}_2'$ – не эффективная оц. ⊗

$\boxed{№4}$



$\xi \sim p(x) = \theta\{(-1;1)\setminus\{0\}\} + (\frac{1}{2}-\theta)\{0\} + (\frac{1}{2}-\theta)\{2\}$

$\vec{x}_n$ $\quad \theta \in (0; \frac{1}{2})$

а) б) $\alpha_1 = M\xi = \int_{-\infty}^{\infty} x\,dF(x,\theta) = \int_{-1}^{1}\theta x\,dx + (\frac{1}{2}-\theta)\cdot 0 + (\frac{1}{2}-\theta)2 = 1-2\theta$

$\alpha_2 = M\xi^2 = \int_{-\infty}^{\infty} x^2\,dF(x,\theta) = \int_{-1}^{1}\theta x^2\,dx + (\frac{1}{2}-\theta)\cdot 0^2 + (\frac{1}{2}-\theta)\cdot 4 = \frac{2}{3}\theta + 2 - 4\theta = 2 - \frac{10}{3}\theta$

$\Rightarrow \mu_2 = M\xi^2 - (M\xi)^2 = 2 - \frac{10}{3}\theta - 1 + 4\theta - 4\theta^2 = 1 + \frac{2}{3}\theta - 4\theta^2$

ОММ: $\alpha_1 = \alpha_1(\theta)$ $\quad 1-2\theta = \bar{x}$ $\quad \tilde{\theta}_1 = \frac{1-\bar{x}}{2}$ – ОММ

несм.: $M\tilde{\theta}_1 = \theta$: $M\tilde{\theta}_1 = M\left[\frac{1-\bar{x}}{2}\right] = \theta \Rightarrow \tilde{\theta}_1$ – несм ✓

сост.: $D\tilde{\theta}_1 = D\frac{1-\bar{x}}{2} = \frac{1}{4}D\bar{x} = \frac{1}{4n}\xi = \frac{1+\frac{2}{3}\theta-4\theta^2}{4n} \xrightarrow[n\to\infty]{} 0$

дост. усл. $\Rightarrow \tilde{\theta}_1$ – сост. ✓

ОМП: $L(\theta) = \prod_{i=1}^{n} p(x_i,\theta)$ Пусть m чисел $\{0, 2\}$ и $n-m$ $(-1;0)\cup(0;1)$

$L(\theta) = \theta^{n-m}(0{,}5-\theta)^m$

$\ln L(\theta) = (n-m)\ln\theta + m\ln(0{,}5-\theta)$

$(\ln L(\theta))' = \frac{n-m}{\theta} + \frac{-m}{0{,}5-\theta} = \frac{(n-m)(0{,}5-\theta)-m\theta}{\theta(0{,}5-\theta)} = \frac{n-m-2n\theta}{\theta-2\theta^2} = 0$

$\Rightarrow \tilde{\theta}_2 = \frac{n-m}{2n} = \frac{1}{2} - \frac{\nu}{2}$ — ОМП.

Проверка на т. max:

$(\ln L(\theta))'' = \frac{-(n-m)}{\theta^2} + \frac{-m}{(0{,}5-\theta)^2} < 0$

$\Rightarrow \tilde{\theta}_2$ — т. макс.

несм.: $M\tilde{\theta}_2 = \theta$: $M\tilde{\theta}_2 = M\left(\frac{1}{2} - \frac{\nu}{2}\right) = \frac{1}{2} - \frac{1}{2}\underset{1-2\theta}{\underset{\parallel}{p}} = \frac{1}{2} - \frac{1}{2} + \frac{2\theta}{2} = \theta$

$\Rightarrow \tilde{\theta}_2$ — несм.

сост.: $D\tilde{\theta}_2 = \frac{1}{4}D\nu = \frac{1(p)(1-p)}{4n} = \frac{(1-2\theta)2\theta}{4n} \xrightarrow[n\to\infty]{} 0$

дост. усл.

$\Rightarrow \tilde{\theta}_2$ — сост.

c) 1) модель — рег.

а) $f(x,\theta) = \theta\{(-1;1)\setminus\{0\}\} + \left(\frac{1}{2}-\theta\right)\{0\} + \left(\frac{1}{2}-\theta\right)\{2\}$ — непр. дифф. по θ на $\theta\in(0,\frac{1}{2})$

б) $\int_{-1}^{1}\frac{\partial f(x,\theta)}{\partial\theta} + \sum p_k(\theta) = 0 \Rightarrow \int_{-1}^{1}dx + -2 = 2-2=0$

в) $I(\theta) = M\left[\left(\frac{\partial \ln f(x)}{\partial\theta}\right)^2\right] = \int_{-1}^{1}\frac{1}{\theta^2}\cdot\theta\,dx + \left(-\frac{1}{\frac{1}{2}-\theta}\right)^2(1-2\theta)$

$= \frac{2}{\theta} + \frac{4}{1-2\theta} = \frac{2}{\theta(1-2\theta)}$ $\Rightarrow I(\theta) > 0$ и непр. на $\theta\in(0;\frac{1}{2})$

$\Rightarrow$ модель рег. (1)

2) оц. — рег. рассм. $\tilde{\theta}_1$

а) модель — рег.

б) оц. — несм.

в) $D\tilde{\theta}_1 = \frac{1+\frac{2}{3}\theta-4\theta^2}{4n}$ — оц. на θ колл. из $(0;\frac{1}{2})$

достат. усл.

$\Rightarrow$ оц. $\tilde{\theta}_1$ — рег. (2)

$\Rightarrow$ из (1) и (2) $\Rightarrow$ работает нер-во К-Р:

$$D\tilde{\theta}_1 = \frac{1+\frac{2}{3}\theta-4\theta^2}{4n} \overset{?}{\geq} \frac{1}{n\left(\frac{2}{\theta(1-2\theta)}\right)}$$

$$1+\frac{2}{3}\theta-4\theta^2 \overset{?}{\geq} 2\theta(1-2\theta)$$

$$1+\frac{2\theta}{3} \overset{?}{\geq} 2\theta \quad \text{т.к. } \theta\in\left(0;\frac{1}{2}\right)$$

$\Rightarrow$ ничего не можем сказать

Рассмотрим $\tilde{\theta}_2 = \frac{1}{2} - \frac{\bar{\theta}}{2}$

а) модель — рег

б) оц. — несм.

в) $D\tilde{\theta}_2 = \frac{(1-2\theta)2\theta}{4n}$ — огр. на $\forall$ компакте из $\theta\in\left(0;\frac{1}{2}\right)$

достат. усл.

$\Rightarrow$ оц. $\tilde{\theta}_2$ — рег. (2')

$\Rightarrow$ из (1) и (2') $\Rightarrow$ работает нер-во К-Р:

$$D\tilde{\theta}_2 = \frac{(1-2\theta)\cancel{2}\theta}{\cancel{2}\cdot 2n} = \frac{1}{n\frac{2}{\theta(1-2\theta)}} = \frac{\theta(1-2\theta)}{2n}$$

достат. усл.

$\Rightarrow$ оц. $\tilde{\theta}_2$ — эффективна

$\Rightarrow$ по (T) ! эфф. оц. $\Rightarrow$ $\tilde{\theta}_1$ — неэфф.